

World-space hand comparison on HOI4D: every method on OUR detector boxes

All metrics in mm, lower is better. *Every* learned row uses our detector boxes, the same 157-sequence held-out set, and a 30-frame WA window, scored with our global metrics. C-MPJPE = *absolute* camera-frame joint error; WA-MPJPE = trajectory error after per-window alignment; W-MPJPE = global trajectory error. FPS is inference throughput (higher is better).

Method	Protocol	C-MPJPE ↓	WA-MPJPE ↓	W-MPJPE ↓	FPS ↑
<i>Offline, metric (per-frame metric model + DROID-SLAM trajectory)</i>					
HaWoR ^b	offline + SLAM	87.7	40.1	53.4	2.47 ^f
HaMeR _{mt} + SLAM ^g	offline + SLAM	67.3	41.8	66.8	2.47 ^f
HaMeR _{ft} + SLAM ^g	offline + SLAM	37.4	36.8	53.6	2.47 ^f
<i>No metric depth (only WA-MPJPE is metric-meaningful)</i>					
WiLoR + SLAM ^a	offline + SLAM	–	43.7	–	2.47 ^f
HaPTIC + SLAM ^c	offline + SLAM	–	36.7	–	2.47 ^f
Dyn-HaMR ^h	offline optim.	–	49.3	–	2.47 ^f
<i>Online / feedforward</i>					
Ours (feedforward) ^e	online, no SLAM	36.2	33.2	37.3	6.59

^a WiLoR + SLAM: per-frame WiLoR cropped from our detector boxes (v3), composed with the same DROID-SLAM trajectory as our HaWoR run. WiLoR produces no metric depth (dummy focal), so its absolute placement is not metric (composed C-MPJPE 1073.7 mm; the pose itself is good, root-relative 32.6). Only the alignment-invariant WA-MPJPE (43.7) is meaningful. CORRECTED 2026-08-06 on two counts. (i) BOX SOURCE: this cell previously read 33.4, which is WiLoR’s *native*-detector score (`wilor_slam_preds`, built from `wilor_native_truefocal_preds`), not a shared-box score, so it was not comparable with the other rows of a table whose premise is one common detector. (ii) JOINT REMAP: the shared-box cache had permuted finger blocks; re-permuting and rescoring the identical 157 sequences moves WA 47.0 → 43.7 and root-relative 55.8 → 32.6 (job 9899520). Vanilla HaMeR + SLAM behaves identically; the metric HaMeR + SLAM rows are the absolute-depth-tuned HaMeR_{mt/ft} above.

^b HaWoR: the full pipeline (DROID-SLAM + Metric3D + motion + infiller) run end-to-end on all 157 sequences with our boxes injected, 0 sequence failures. On its *own* tuned detector it scores 49.7 / 34.7 / 42.3; on our (noisier) boxes it degrades to 87.7 / 40.1 / 53.4. This is the fair, input-matched number: HaWoR’s motion+infiller is tuned to its own box statistics, so under a common realistic detector our jitter-robust model holds up while HaWoR does not. ^c HaPTIC + SLAM: HaPTIC produces no metric depth, so absolute placement is not recoverable (C-MPJPE 157.7 with a depth-dependent error, while root-relative C-MPJPE is a healthy 29.2 – the pose is good, only metric placement fails). HaPTIC does no camera tracking, so we compose its per-frame camera-space hand with the *same* DROID-SLAM trajectory as the other rows (SE(3) residual 0.000 mm) and score identically. Only WA-MPJPE is meaningful: 36.7 on our detector boxes. BUILT 2026-08-06 (Euler 9906263): this cell previously carried 35.7, which is HaPTIC’s *native*-regime score, because no shared-box composition existed. It does now (`haptic_slam_detbox_preds`), and the two agree to 1.0 mm, as expected for a method whose world error is dominated by the shared trajectory rather than by the boxes. A third variant on disk, `haptic_detbox_preds`, scores 23.3 and must NOT be used: its world track is lifted with ground-truth extrinsics, so it is an oracle. ^e All of our numbers come from a single end-to-end feedforward run (our boxes, no SLAM, predicted extrinsics, 30-frame WA window); the head is trained jitter-robust, so it tolerates the same noisy boxes that degrade the offline baselines. Feedforward FPS = 6.59, measured on an NVIDIA GB10 (mean over 30 timed 16-frame clips; a single 30-frame forward pass is 4.5 fps). Total params 1178.7 M, trainable 262.4 M. ^f Offline FPS is MEASURED: the full HaWoR pipeline (detect + DROID-SLAM + Metric3D + motion + infiller) ran 157 sequences of 300-frame video in 5 h 18 min = 2.47 fps, dominated by offline DROID-SLAM. The load-bearing distinction is *online/feedforward* (single causal pass, no future frames, no global optimization) vs *offline* (whole-video SLAM), not raw throughput. ^g HaMeR_{mt}/HaMeR_{ft}: HaMeR fine-tuned on HOI4D with our absolute-depth (kp3d-abs) recipe (mt = metric-tuned translation head; ft = stronger fine-tune), per-frame on our boxes, composed with the same DROID-SLAM trajectory. Our recipe transfers to a HaMeR backbone (HaMeR_{ft}+SLAM reaches C-MPJPE 37.4, on par with our feedforward 36.2) – but they remain *offline* (need whole-video SLAM), whereas ours is a single online pass. ^h Dyn-HaMR input-matched: its per-frame estimator seeded from our boxes (WiLoR init) and its DROID-SLAM stage replaced by our SLAM cameras; PHALP and DROID are bypassed. Dyn-HaMR is an *optimization* method (per-sequence test-time fitting). Because the input-matched setup omits its own metric-depth stage, its world-scale is unconstrained and collapses on ego HOI4D: absolute C-MPJPE and W-MPJPE blow up (1337 / 79 mm) and are omitted. Only the scale-invariant WA-MPJPE (49.3) is meaningful. Reported over 156 of 157 sequences.